

## **ABSTRACT**

AI-Based Virtual Assistant to enhance your communication skills is an Intelligent System with state of art of AI technologies like NLP, Speech recognition, ML which help the user to speak better in English language. The system takes voice input by listening continuously the users spoken words and converts them into text with the help of speech-to-text service such as Amazon Polly, allowing users to hear the correct pronunciation while reading the improved captions. A unique feature of the proposed system is its English-Only Mode, which utilizes language detection techniques to monitor the user's speech throughout the conversation.. This will display as real time live captions of user's speech on the screen. Then this text input will be passed to Large Language Model (LLM) to identify grammar errors, improvement of vocabulary, structure of the sentences and give some suggestion to speak like in real world like professional, more natural than current one. This will then be shown as revised answer in screen and again converted into voice with text-to-speech service (e.g., Amazon Polly) to listen the corrected pronounce while reading the caption in screen. Unique English-Only Mode feature is enabled by detecting user's voice language at runtime with language detection, so if user change to some other local language or mixes both language, automatically it give a voice & text response, such as, "Please use only English". The speech in both text and voice, pronunciation analysis, confidence level, fluency check, pace of speaking, and removal of the unnecessary fillers (um and uh) will be performed by the AI System and it will generate the feedbacks for improvement in user's communication skill. This also contains an AI speaking partner to practice conversational speak. Other feature includes of a tool which helps to conduct practice interviews, simulation for group discussions and daily speaking and vocabular challenges to build up user's confidence on communication. The platform also contains personalized progress dashboard to record and track user performance of the past days to see improvements in grammar, vocab, pronunciation, fluency, confidence, etc. By utilizing the capabilities of Real-time speech recognition, advanced AI Language correction, unique English-Only enforcement, speech synthesis, and in-depth performance tracking, this system provides a complete, effective, and engaging platform for students, job seekers, professionals and every learner to boost their English communication skill and achieve their goals in a more confident way.

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